Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Enterprise AI Agents · Complex Domain Workflows · Production Python · Data Pipelines · Multi-Tenant SaaS · Full-Stack · AWS

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SUMMARY

Full-stack software engineer with 9+ years building enterprise AI agent systems, data pipelines, and product interfaces end-to-end - from infrastructure through UI. Currently at Intuit integrating Anthropic Claude into production financial services: **autonomous AI workflows**, tool integrations, structured outputs, guardrails, and evaluation infrastructure serving Fortune 500-scale consumers. Built **Leasel** - a production agentic platform using LangGraph (8-agent orchestration) for **complex domain-specific workflows**: tool calling, durable state, MCP server integration, hybrid search RAG, HITL guardrails, and a CI evaluation harness (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLM evaluation platform with cost-aware multi-provider routing, CI regression gating, and full observability. At Entrata (7+ years), owned **mission-critical data pipelines** and async distributed systems in a regulated multi-tenant SaaS serving 100K+ enterprise customers - HUD compliance ETL, RabbitMQ event-driven processing with idempotency guarantees, PHI-compliant S3 migration. Strong full-stack: Python/FastAPI backend + React/TypeScript frontend. Brings **independent technical judgment** and end-to-end ownership from architecture decisions through production operations. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Enterprise AI Agents · Complex Domain Workflows · Orchestration | Multi-Tenant Systems · Enterprise SaaS · Regulated Environments | Production Python · FastAPI · Distributed Services | Data Pipelines · Messy Enterprise Data · ETL · Integration | Infrastructure through UI · Full-Stack Engineering | LangChain · LangGraph · Agentic Workflows · Tool Use | LLM Evaluation · LLM-as-Judge · Guardrails · Observability | Independent Technical Judgment · End-to-End Ownership | CTO Collaboration · Architecture · System Design | React · TypeScript · Node.js · Frontend Interfaces | AWS (Lambda · Bedrock · SQS · API Gateway) · Docker · Kubernetes | Async Processing · RabbitMQ · Event-Driven · Idempotency | Fortune 500 Enterprise · Customer-Facing Product · Remote-First

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **Enterprise AI agent systems:** Design and build LangChain-based agentic workflows integrating Anthropic Claude into production financial services for Fortune 500-scale consumers - tool calling, retrieval-grounded context management, structured outputs, and third-party platform integrations; guardrails with input validation, sensitive-data handling, and fallbacks enforcing reliable production behavior on complex domain workflows.
- **Evaluation and observability:** Build LLM-as-judge evaluation harnesses with golden datasets and CI regression gates; monitor production AI workflows via structured logging and CloudWatch; perform qualitative error analysis and turn failures into eval coverage. Document patterns and playbooks for team reuse.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS with full infrastructure ownership from data layer through product interface.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant enterprise, 100K+ monthly active users)

- **Mission-critical data pipelines and messy enterprise data:** Owned HUD income-limit compliance ETL end-to-end (180K records, zero violations across 6 years); RabbitMQ event-driven async processing with idempotency guarantees and dead-letter routing for rent updates across thousands of enterprise tenant records; PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- **Multi-tenant enterprise SaaS ownership:** Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; exercised independent technical judgment on architecture decisions; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Enterprise AI Agent Platform: Complex Domain Workflows · LangGraph · MCP · RAG

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production enterprise AI agent platform handling **complex, document-heavy domain workflows**: **8-agent LangGraph orchestration** with tool/function calling, durable state, conditional routing; **MCP server integration** for shared tool access; **hybrid search RAG** (OCR ingestion, chunking, TF-IDF + semantic reranking, citation grounding) over messy enterprise documents; **HITL guardrails** for regulated decisions; **CI eval harness** (LLM-as-judge, golden datasets: 93.6% precision, 1,240 test cases). Three product surfaces share one data and retrieval layer - infrastructure built for re-use.

AgentProbe - LLM Evaluation & Observability Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source eval and observability platform: offline golden-dataset evals, LLM-as-judge scoring, CI regression gating, failure handling with structured error classification; cost-aware routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full tracing and structured logging per run. Used as a production release gate before every AI deploy.

AI on FHIR — Clinical NLP & Enterprise API Integration 2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. Third-party platform integration, OAuth-secured APIs, HIPAA-style RBAC, Docker-containerized. Full-stack: Python/Flask backend + React frontend.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017